

Executive Summary:

Early Ego-Network Openness and Cross-Field Breadth of Scientific Concepts

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Headline

Scientific concepts whose early topic co-occurrence networks are more open—more diverse partners, more partner turnover, lower clustering—go on to spread across more disciplines. The $\text{OPEN}_{\text{home}}$ index, a composite of six ego-network indicators measured from home-field papers only, shows a partial Spearman correlation of +0.069 [+0.038, +0.100] with rarefied cross-field breadth, pooled over six non-selection bodies (four held-out field groups, a 2010–2014 onset cohort, and a fresh 2015–2017 onset cohort), with $I^2 = 0$ and 6/6 bodies positive. The association is modest (no practical forecasting gain over simple popularity features) but robust across fields and concept populations.

Goal

Determine whether early structural signals in a scientific concept’s topic co-occurrence network anticipate later cross-field breadth, beyond simple measures of volume and reach. The study uses the OpenAlex corpus (476 million works), tracking 12,499 concepts and 27,393 adoption episodes across 26 fields over a five-iteration research run.

What was tried

1. **Iteration 1** screened five rival mechanisms (lineage naturalisation, co-author reach, co-occurrence diversity, frequency-free selectivity, gateway landing) on a 48-concept dev panel. None survived the pre-registered rule.
2. **Iteration 2** scaled up to a 12,499-concept panel. Gateway centrality for retention was disconfirmed ($\Delta\text{AUC} = -0.00001$). Retained-field relatedness predicts field entry ($d_0 = 0.302$).
3. **Iteration 3** ran the held-out indicator screen: 53 indicators, 10 frozen on DEV, 7 confirmed on held-out groups spanning two families (retained frontier and ego-network topology). Also ran the decisive retained-frontier test ($d_0 = 0.322$ on independent frame).
4. **Iteration 4** confirmed OPEN on a fresh 2015–2017 cohort (marginal: +0.080 at R3, Holm $p = 0.048$). Breadth decomposition: 73% exploration, 27% retention. Within-concept closure test: NOT SUPPORTED.
5. **Iteration 5** confirmed the signal on vocabulary-free “Frame N” concepts (+0.117 at R3); showed that Cheng’s consistency predicts volume but *narrower* reach (−0.069); found the signal is topical non-redundancy (static partner diversity), not year-to-year churn.

Key results

- Seven of 53 early network indicators survive held-out validation for rarefied breadth. They span two families: retained frontier (M0_density_end, D_vol_end, CONTACT_REACH, RETENTION_RATIO) and ego-network (n_comm, NOV_{res}, ego_density).
- OPEN_{home} PSP +0.069 [+0.038, +0.100] over 6 non-selection bodies ($I^2 = 0$).
- Retained-field relatedness predicts field entry: $d_0 = 0.322$ [0.291, 0.355] on 6,978 entry events (FRONTIER = PARTIAL; backbone-specific).
- The consistency–breadth reversal: ideational consistency predicts volume (+53.5%/SD) but narrower reach (−0.069; all five field groups negative, $I^2 = 0$).
- Breadth decomposition: early contact diversity accounts for 73% of the gap; retention 27%; frontier advance near zero.
- The signal is topical non-redundancy of the home-topic partner mix, not year-to-year partner turnover (within-concept permutation excess +0.008 ≈ 0).

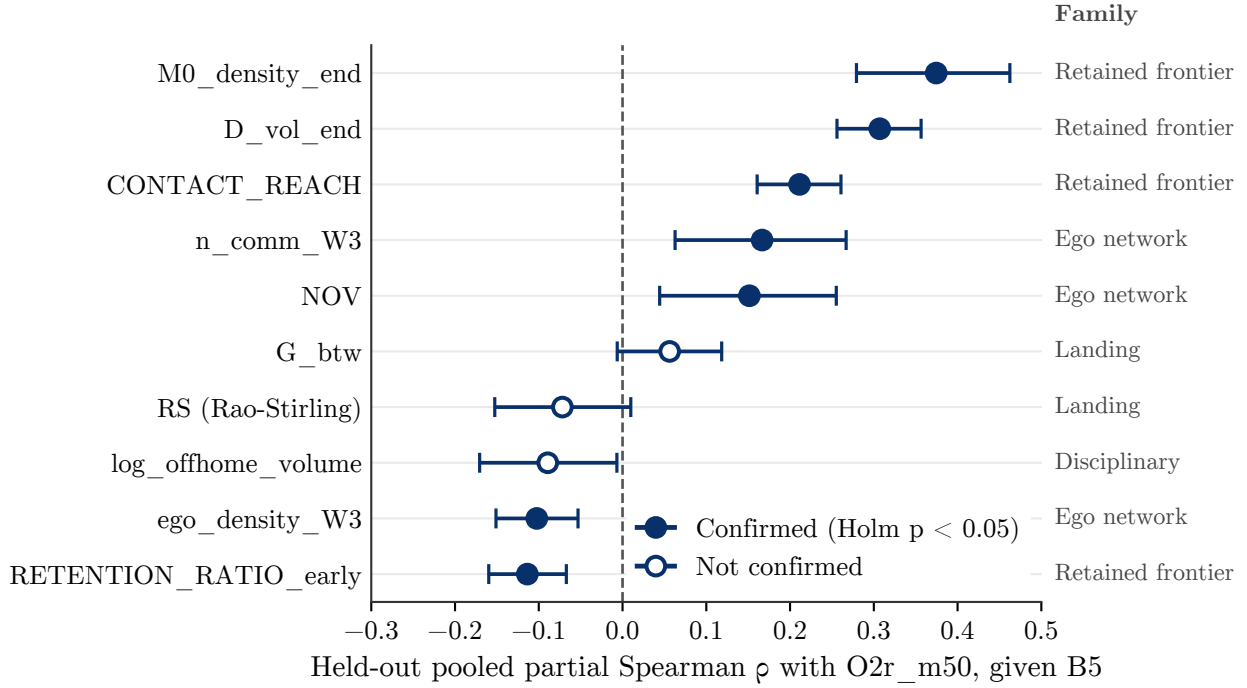


Figure 1: Held-out screen of the 10 indicators frozen on DEV for rarefied breadth. Seven are confirmed (filled circles, Holm $p < 0.05$): five positive, two negative.

What failed

- A_h^* naturalisation gap, D_{ratio} , gateway landing G : fail on dev screen (iteration 1).
- Gateway centrality for retention: $\Delta AUC = -0.00001$ on 27,393 episodes (iteration 2).
- Trajectory typology: no discrete types pass the naming rule ($ARI = 0.222$).
- Within-concept closure (Experiment 11): NOT SUPPORTED; opposite sign on held-out.
- OPEN_{home} forecasting gain: +0.002 over baseline—no practical prediction.
- Four artifacts failed or were incomplete across the run.

Open questions

- The persistence-filtered RCA rival ($D_{\text{rca_persist.k}}$) remains untested.
- LIFEENV weakness (PSP +0.071 vs +0.186 for other bodies): unexplained.
- Causal mechanism: the association is correlational—openness could reflect intrinsic generality, community diversity, or problem breadth.
- Domain heterogeneity in field entry ($I^2 = 0.92$).
- Degree-normalised ego density and Palla size $\times$ turnover interaction were recommended but not run.